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Editor-in-Chief
Foundations of Physics
Springer Nature

Subject: Submission of Manuscript "Fractal Information Nadsoliton (FIN) Theory: From Quantum Foam to Cosmic Web via Network Dynamics"

Dear Editor,

I am pleased to submit the attached manuscript, "**Fractal Information Nadsoliton (FIN) Theory: From Quantum Foam to Cosmic Web via Network Dynamics**," for consideration in *Foundations of Physics*.

Core Contribution:

This paper introduces a **Master Equation** for spacetime dynamics derived from a single, zero-parameter coupling kernel defined on a fractal information network. By treating the universe as a self-learning system, we demonstrate that fundamental laws emerge from network plasticity:

- **Gravity** emerges as **Hebbian Learning** (network connections strengthen with activity, contracting effective distance).
- **Dark Energy** emerges as **Network Forgetting** (unused connections decay, driving expansion).
- **The Hierarchy Problem** (10^{40} gap) is resolved through **Fractal Scaling** across 20 layers of self-similarity.

Appropriateness for *Foundations of Physics*:

This work addresses the deepest foundational questions: the origin of physical constants, the nature of spacetime, and the mechanism of unification. Unlike standard approaches that introduce arbitrary parameters, our framework derives constants such as the **Fine Structure Constant** ($\alpha^{-1} \approx 137.115$) and the **Proton Radius** (0.84 fm) from pure information geometry with high precision and **zero fitting**.

We believe this novel perspective—viewing General Relativity as the hydrodynamics of a learning network—offers the conceptual breakthrough required to bridge the gap between quantum information and gravity.

All results are backed by over 490 independent simulations. The full codebase ensuring reproducibility is available at:

<https://github.com/hyconiek/Fractal-Nadsoliton-Theory>.

Thank you for your consideration.

Sincerely,

Krzysztof Żuchowski
Independent Researcher